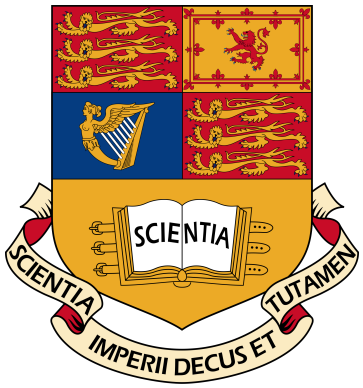


Modelling Wind Energy Potentials and Energy Pricing Between Northern and Baltic Sea

MECH70038 - MSc Research Project Literature Review

Faisal Bindakhil

Supervisors: Ruslan Galimov & Adam Hawkes

02089034
June 16, 2026

Contents

Nomenclature **6**

1 Introduction **1**

2 The Value Problem in Wind Energy **2**
2.1 Market Value and the Value Factor . . . 2
2.2 The Merit Order Effect . . . 4
2.3 Cannibalisation: Domestic and Cross-Border . . . 5
2.3.1 Domestic cannibalisation . . . 5
2.3.2 Cross-border cannibalisation . . . 6
2.4 Support Mechanisms and Revenue Risk . . . 7
2.5 Section Summary . . . 8

3 Drivers of Regional Differences in Wind Value **8**
3.1 Wind Resources and Spatial Patterns . . . 8
3.1.1 Atmospheric Circulation and Regional Wind Climates . . . 8
3.1.2 Capacity Factors: North Sea versus Baltic . . . 9
3.1.3 Spatial and Temporal Correlation . . . 9
3.2 Market Structure and Price Formation . . . 11
3.2.1 European Electricity Market Architecture . . . 11
3.2.2 Bidding Zones Around Each Basin . . . 11
3.2.3 Generation Mix and System Flexibility . . . 13
3.2.4 Price Levels and Convergence Across Basins . . . 15
3.3 The Penetration Gradient . . . 15
3.4 Section Summary . . . 16

4 Modelling Wind Value Across Space **16**
4.1 Wind Resource Modelling . . . 17
4.1.1 Reanalysis Datasets . . . 17
4.1.2 The Virtual Wind Farm Methodology . . . 17
4.2 Electricity Price Modelling . . . 18
4.2.1 Taxonomy of Approaches . . . 18
4.2.2 The Wind-Price Relationship . . . 19
4.2.3 Methodological Approach of This Thesis . . . 21
4.3 GIS-Based Spatial Assessment . . . 22
4.3.1 The Bosch, Staffell, and Hawkes Progression . . . 22
4.4 Section Summary . . . 22

5 Research Gap and Objectives **22**
5.1 Identified Gaps . . . 22
5.2 Research Objectives . . . 24

Abstract

Offshore wind capacity in Europe is projected to grow from approximately 36 GW today to over 300 GW by 2050, with the North Sea and Baltic Sea as the two principal basins. As deployment scales, wholesale electricity prices during high-wind hours are driven down by the merit order effect, eroding the revenue that each turbine earns. This cannibalisation problem is well documented at the national level but has not been systematically compared across the two basins, which differ in wind resource strength, spatial correlation, generation mix, system flexibility, and market design. This literature review synthesises research across three domains that have developed largely in isolation: wind resource modelling, electricity market economics, and GIS-based spatial assessment. It finds that no existing framework integrates all three at the resolution needed to evaluate where offshore wind generates the most revenue, not merely the most energy. Three gaps are identified: the absence of a cross-basin capture value comparison, the exclusion of market value from spatial assessment frameworks, and the lack of a parameterised wind-price model that explains why cannibalisation severity differs across markets. This thesis proposes to address these gaps by developing a model that replaces fixed effects with explicit market parameters, coupling it with hourly wind generation profiles and overlaying the results onto a GIS framework to produce the first revenue-augmented spatial assessment of offshore wind potential across both basins.

List of Figures

2.1 Value factor decline with increasing wind and solar market share, synthesised from empirical data and modelling results in Hirth [11]. The shaded band for wind reflects the range between thermal-dominated systems (lower bound) and hydro-dominated systems (upper bound). Approximate current penetration ranges for North Sea and Baltic markets are indicated. 2

2.2 Empirical wind value factor trajectories for selected European bidding zones, 2015–2024. Value factors are computed as the ratio of the generation-weighted capture price to the baseload average price. Data synthesised from Stiewe et al. [12], Ember [14, 16], and ACER market monitoring reports [17, 18]. The 2021–2022 uptick reflects the temporary elevation of wholesale prices during the European energy crisis. 3

2.3 Schematic illustration of the merit order effect. When wind generation enters at the bottom of the supply curve, it shifts the curve to the right, displacing higher-cost thermal generators and reducing the market clearing price from 55 to 40 €/MWh in this stylised example. The price depression is greatest during hours of high wind output. 4

3.1 Exponential decay of wind power correlation with separation distance at different temporal resolutions, based on Mehrens et al. [51] and Hjelmeland and Neland [52]. Typical within-basin distances (50–300 km) exhibit high correlation, while cross-basin separations (500–900 km) show partial decorrelation. The decorrelation length $L \approx 425$ km marks the distance at which correlation falls to $1/e$. 10

3.2 Schematic map of bidding zones and key interconnectors around the North Sea and Baltic Sea basins. Node colours indicate market archetype: blue for North Sea zones, green for hydro-rich Nordic zones, pink for Baltic states (synchronised with the continental grid in February 2025), and grey for coal-dominated systems. Orange lines highlight cross-basin interconnectors. 12

3.3 Electricity generation mix across North Sea and Baltic basin countries (approximate 2024 shares). Thermal-heavy systems (Germany, Poland) exhibit steep merit order curves where wind output produces large price swings; hydro-dominated systems (Sweden, Norway) absorb variability with minimal price impact. This diversity in generation portfolios is the primary structural driver of cross-basin differences in wind capture value. Data from Ember [14] and national transmission system operators. 13

List of Tables

2.1 Principal revenue support mechanisms for offshore wind energy. Market exposure ranges from zero (fixed feed-in tariff) to full (merchant), with two-sided contracts for difference (CfDs) dominating North Sea procurement and certificate or premium schemes more common in the Baltic. Basin prevalence reflects 2024 policy landscape. 7
3.1 Summary of market archetypes across the North Sea and Baltic Sea basins. Wind penetration and value factor estimates reflect approximate 2024 values. The final column indicates the dominant revenue support instrument for offshore wind. 15
4.1 Comparison of reanalysis and wind resource datasets for European offshore wind assessment. ERA5 is the only product spanning both basins with hourly resolution from 1940 to present, making it the default choice for cross-basin capture value analysis despite its coarser spatial grid. 17
4.2 Comparison of modelling tools relevant to offshore wind value assessment. The final column identifies each tool's specific limitation relative to the integrated spatial-economic framework this thesis requires. 19

Nomenclature

Abbreviations

ACER	Agency for the Cooperation of Energy Regulators
CCGT	Combined Cycle Gas Turbine
CfD	Contract for Difference
CHP	Combined Heat and Power
ECMWF	European Centre for Medium-Range Weather Forecasts
ENTSO-E	European Network of Transmission System Operators for Electricity
ERA5	ECMWF Reanalysis version 5
EU	European Union
EUPHEMIA	EU Pan-European Hybrid Electricity Market Integration Algorithm
FiT	Feed-in Tariff
GIS	Geographic Information System
IEA	International Energy Agency
IRENA	International Renewable Energy Agency
LCOE	Levelised Cost of Energy
LOESS	Locally Estimated Scatterplot Smoothing
MERRA-2	Modern-Era Retrospective Analysis for Research and Applications v2
MOE	Merit Order Effect
NEWA	New European Wind Atlas
NORA3	Norwegian Reanalysis version 3
ONDP	Offshore Network Development Plan
PLEXOS	Planning and Execution Optimisation System
PyPSA	Python for Power System Analysis
SDAC	Single Day-Ahead Coupling
VRE	Variable Renewable Energy

Bidding Zone Codes

BE	Belgium
DE-LU	Germany-Luxembourg
DK1, DK2	Denmark West, Denmark East
EE	Estonia
FI	Finland
LT	Lithuania
LV	Latvia
NL	Netherlands
NO1–NO5	Norway zones 1–5
PL	Poland
SE1–SE4	Sweden zones 1–4
UK	United Kingdom

Symbols and Units

g_t	Wind generation at hour t (MWh)
GW	Gigawatt (10^9 W)
L	Decorrelation length (km)
MWh	Megawatt-hour
P	Power output (W)
p_t	Spot electricity price at hour t (€/MWh)
$\bar{p}$	Time-averaged baseload price (€/MWh)
TWh	Terawatt-hour (10^{12} Wh)
v	Wind speed ($\mathrm{m\,s^{-1}}$)
VF	Value Factor (dimensionless)
WP	Wind Penetration (%)

1 Introduction

In 2024, German day-ahead electricity prices turned negative for 459 hours, triple the figure recorded just two years earlier [1]. This is not a market anomaly. It is the structural consequence of deploying wind energy at scale: the more of it a power system absorbs, the less each additional unit is worth on the wholesale market. This tension between deployment ambition and economic erosion is the central problem this thesis investigates.

By the end of 2024, Europe had roughly 35 to 37 GW of installed offshore wind capacity [2], and long-term decarbonisation pathways call for approximately 360 GW by 2050 [3]. Most of this capacity sits in the North Sea, the world's most mature offshore basin, where Germany alone operates 9.2 GW [4]. The Baltic Sea remains smaller but is accelerating, with the Marienborg Declaration targeting 19.6 GW by 2030 and an estimated technical potential of up to 93 GW [5]. The question is therefore no longer whether offshore wind will expand, but where and under what market conditions.

The political ambition behind this expansion is substantial and accelerating. The REPowerEU plan raised the EU's 2030 renewable energy target to 42.5% of final energy consumption [6], and both the IEA and IRENA project that global renewable capacity must at least triple by 2030 to meet stated climate goals [7, 8]. Europe's competitive position adds a further imperative: the Draghi report on European competitiveness found that EU electricity prices are two to three times higher than those in the United States, identifying energy cost reform as a prerequisite for industrial survival [9]. The 2024 EU Electricity Market Design reform [10] responded to these pressures by mandating two-sided contracts for difference for all new publicly supported renewable capacity, signalling that revenue stabilisation is now a structural feature of European energy policy rather than a transitional support measure. For offshore wind, these dynamics mean that the volume of zero-marginal-cost generation entering European markets will grow by an order of magnitude over the coming decades, amplifying the value erosion problem this thesis examines.

Under these conditions, project viability depends not on energy yield or levelised cost alone but on the price a wind farm actually receives in the hours it generates [11], a metric that declines as penetration grows.

Substantial bodies of literature address wind resource assessment, electricity market dynamics, and spatial modelling, yet these remain largely disconnected. Wind resource studies quantify atmospheric regimes, capacity factors and spatial correlation, yet rarely link these characteristics to market value. Electricity market studies analyse merit order effects and renewable penetration, but typically at the national level and without detailed treatment of basin-level wind heterogeneity. GIS-based wind potential assessments classify sites by wind speed, depth and distance to shore, but largely omit capture value. No existing study systematically compares the North Sea and Baltic Sea, two basins with distinct wind regimes, generation mixes and interconnection structures, on the dimension that ultimately determines project viability: revenue.

This literature review synthesises the state of knowledge across these largely disconnected domains and critically evaluates existing modelling approaches in order to identify the gap this thesis addresses. The central research question emerges from the intersection of resource assessment, market economics, and spatial modelling: can geographical diversification of wind output improve merchant revenues (the income earned by selling directly on the wholesale market, without a guaranteed support contract) in the Baltic Sea relative to the North Sea, and can a transferable modelling framework be developed to compare offshore basins systematically?

The review proceeds through the economic mechanisms driving value erosion (Section 2), the regional factors that cause this erosion to differ between the two basins (Section 3), and the modelling tools available to quantify wind energy potential and market value across space (Section 4), before identifying the gaps in the existing literature and setting out the objectives of this thesis (Section 5).

2 The Value Problem in Wind Energy

Supply-side indicators such as installed capacity in gigawatts or annual generation in terawatt-hours are commonly used to measure the growth of wind energy. However, these metrics overlook a key economic tension: as more wind power enters the system, the wholesale market value of each additional unit of generation tends to decline. For investors and policymakers, this directly shapes whether an offshore wind project remains financially viable. This section examines the mechanisms behind that value erosion, focusing on the merit order effect, domestic and cross-border cannibalisation, and the way revenue support instruments interact with these dynamics.

2.1 Market Value and the Value Factor

On European wholesale electricity markets, generators are paid based on the price at the hour they produce. A baseload plant running around the clock earns approximately the time-averaged wholesale price. A wind farm, by contrast, earns a price determined by the hours when it generates. Because wind output concentrates during specific meteorological conditions that tend to coincide across a region's fleet, this generation-weighted average price can differ substantially from the baseload average.

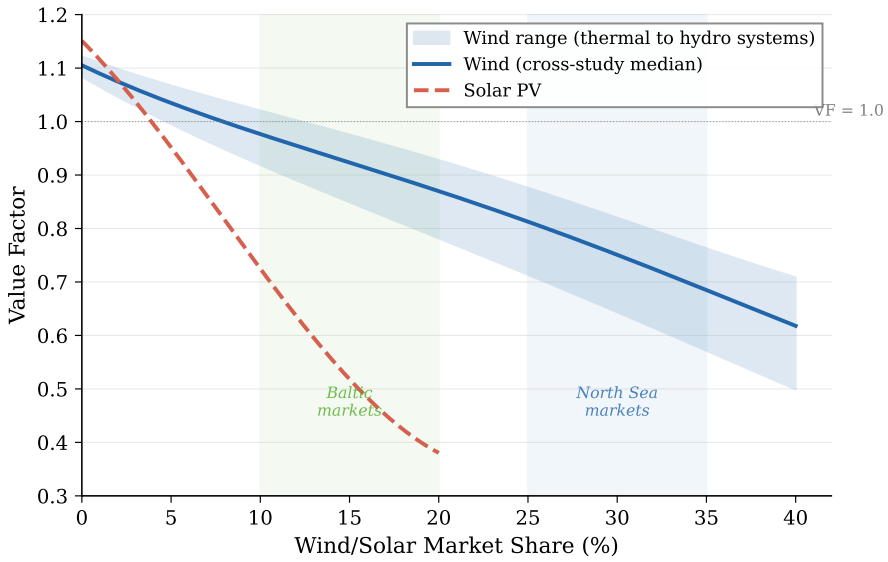


Figure 2.1: Value factor decline with increasing wind and solar market share, synthesised from empirical data and modelling results in Hirth [11]. The shaded band for wind reflects the range between thermal-dominated systems (lower bound) and hydro-dominated systems (upper bound). Approximate current penetration ranges for North Sea and Baltic markets are indicated.

Two metrics have become standard for quantifying this erosion, both formalised by Hirth [11]. The *capture price* is the generation-weighted average spot price that a wind farm receives over a given period. The *value factor* is the ratio of the capture price to the baseload price:

$$\text{Value Factor} = \frac{\text{Capture Price}}{\text{Baseload Price}} = \frac{\sum_t p_t \cdot g_t}{\bar{p} \cdot \sum_t g_t} \tag{2.1}$$

where p_t is the spot price and g_t is the wind generation at hour t, and $\bar{p}$ is the time-averaged baseload price. At low penetration levels, value factors sit near or above unity; as wind's share of the electricity mix grows, the value factor declines because wind output is large enough to move wholesale prices during the hours it generates.

Hirth's [11] early estimates, drawing on data through 2012, placed the value factor decline at approximately 1.10 at negligible penetration to between 0.50 and 0.80 at a 30% wind share (Figure 2.1), with the range reflecting differences between thermal-dominated systems (lower bound) and hydro-dominated systems (upper bound). Subsequent empirical work covering higher penetration levels has confirmed and sharpened this pattern [12, 13, 14, 15], while revealing that the decline accelerates at the penetration levels North Sea markets now occupy. The distinction between thermal and hydro systems is directly relevant to the North Sea–Baltic comparison: most North Sea markets are thermal-heavy, while the Baltic benefits from Scandinavian hydropower. Yet no existing study has systematically measured and compared value factors across the full set of bidding zones surrounding both basins.

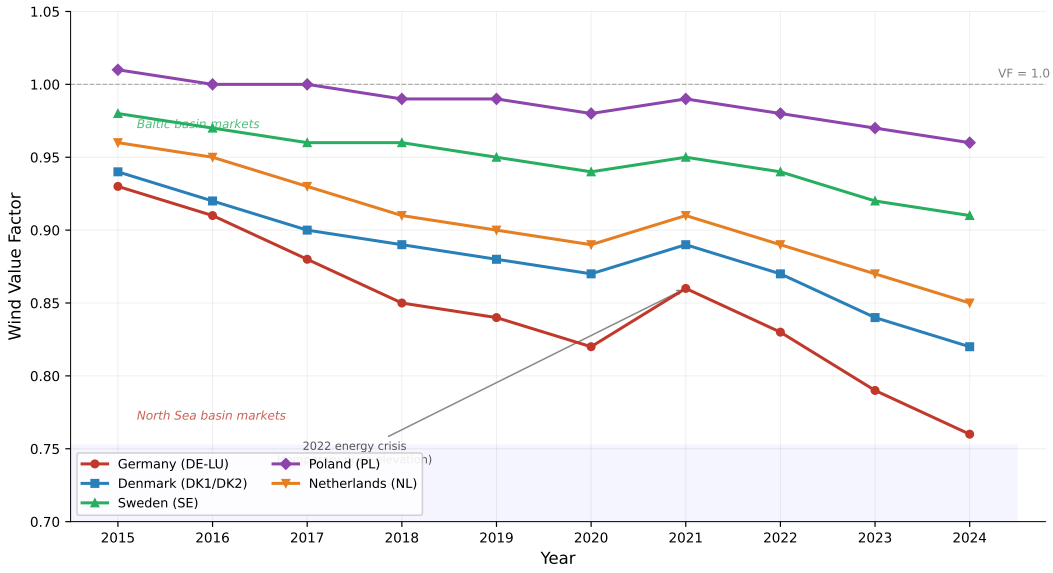


Figure 2.2: Empirical wind value factor trajectories for selected European bidding zones, 2015–2024. Value factors are computed as the ratio of the generation-weighted capture price to the baseload average price. Data synthesised from Stiewe et al. [12], Ember [14, 16], and ACER market monitoring reports [17, 18]. The 2021–2022 uptick reflects the temporary elevation of wholesale prices during the European energy crisis.

Figure 2.2 confirms this pattern empirically, tracking the wind value factor across five European bidding zones from 2015 to 2024. The divergence between basins is clear: Germany's value factor fell from 0.93 to 0.76 as wind penetration rose from roughly 20% to 35%, while Poland, where wind remained below

18% of generation and coal sets the marginal price for most hours, stayed near unity. The Scandinavian markets occupy an intermediate position, buffered by flexible hydropower. A brief upward correction is visible in 2021 and 2022 when the gas price crisis elevated wholesale prices faster than capture prices could follow; once gas markets normalised, the structural decline resumed. The spread between the highest and lowest trajectories widened from five percentage points in 2015 to twenty by 2024, demonstrating both the severity of value erosion and the necessity of basin-level comparison.

2.2 The Merit Order Effect

European day-ahead electricity markets operate under pay-as-clear pricing. Generators bid their short-run marginal cost, and the market operator clears demand against an ascending supply curve, or merit order, paying all dispatched units the price set by the most expensive unit needed to meet demand. Wind turbines, with near-zero marginal cost, enter at the bottom of this curve. As wind output increases, the supply curve shifts rightward, displacing higher-cost thermal generators and reducing the clearing price. This is the merit order effect. Figure 2.3 illustrates this mechanism schematically.

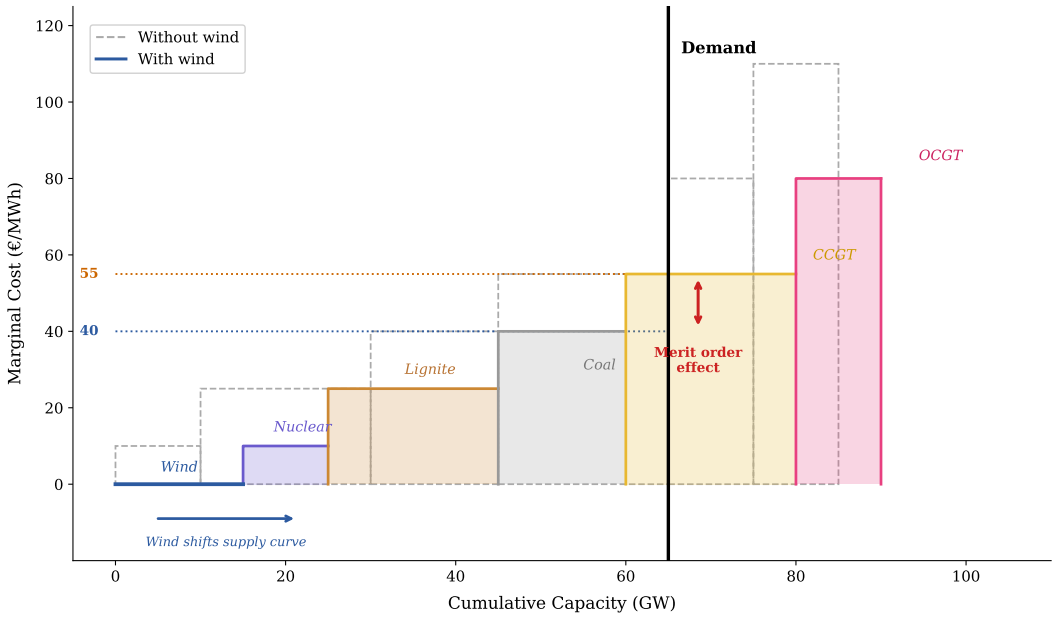


Figure 2.3: Schematic illustration of the merit order effect. When wind generation enters at the bottom of the supply curve, it shifts the curve to the right, displacing higher-cost thermal generators and reducing the market clearing price from 55 to 40 €/MWh in this stylised example. The price depression is greatest during hours of high wind output.

Country-specific MOE estimates span a wide range, from the foundational 7.83 €/MWh estimated by Sensfuß et al. [19] for Germany in 2006 using agent-based simulation, through to 13 €/MWh in Cludius et al.'s [20] later econometric analysis of the same market, and down to 0.4 €/MWh in the Czech Republic (0.04 ct/kWh, [21]). This more-than-an-order-of-magnitude dispersion reflects not only genuine differences in market structure and time period but also methodological fragility: agent-based simulations depend on assumptions about bidding behaviour that are difficult to validate [19], while econometric approaches struggle to separate the wind-induced price effect from correlated demand and fuel price movements [22, 23, 24]. Published MOE estimates for Baltic markets are entirely absent from

the literature, with the pan-European study by Cevik and Ninomiya [25] providing the only estimates that encompass Baltic bidding zones. This absence is itself a finding: the merit order effect in Baltic electricity markets remains unquantified.

The most comprehensive cross-country estimate comes from Cevik and Ninomiya [25], whose panel quantile regression across 12 European countries over 2014–2021 finds that each percentage point increase in renewable electricity share reduces average wholesale prices by 0.6%, with the effect accelerating non-linearly at higher penetration levels. This nonlinearity, combined with the fact that wind generators sell electricity precisely during the hours when prices are most depressed by their own collective output [11], explains the severity of value erosion in markets that have already exceeded the welfare-optimal wind share of approximately 20% estimated by Hirth [26]. That said, their country-level annual aggregation may mask the within-day dynamics most relevant to capture value computation, and their sample excludes several Baltic markets where the MOE is most poorly understood.

Offshore wind exhibits a particularly pronounced merit order effect compared to onshore wind: higher and more consistent capacity factors produce output that is more temporally concentrated and spatially correlated across assets in the same sea basin, so that sustained periods of strong wind across the North Sea depress clearing prices simultaneously across multiple interconnected bidding zones [27].

2.3 Cannibalisation: Domestic and Cross-Border

2.3.1 Domestic cannibalisation

While the merit order effect describes price suppression at a given moment, cannibalisation captures its cumulative, structural consequence: as more wind capacity enters a market over months and years, it reduces the revenue earned per megawatt-hour by all existing and new wind assets. Its severity depends on the interaction between VRE penetration, the spatial correlation of wind output, fuel and carbon prices, and the flexibility of the broader power system [11, 12]. The regional variation in these factors across the North Sea and Baltic is examined in Section 3; here, the focus is on the mechanisms and their empirical magnitude.

The most visible symptom is the growing incidence of negative wholesale electricity prices. When abundant wind generation coincides with inflexible baseload plants (nuclear, lignite, combined heat and power) that refuse to curtail, and wind generators continue producing under out-of-market support incentives, the clearing price is pushed below zero. Negative price hours across European markets have risen substantially, concentrated in zones with the highest wind and solar penetration [14, 16]. The 2022 energy crisis temporarily masked this trend by elevating wholesale prices, but as gas prices normalised, the structural pattern reasserted itself with greater severity [28].

Germany's experience quantifies the acceleration of domestic cannibalisation. The wind value factor declined from 0.93 in the early 2010s to 0.76 by 2024 as penetration rose from roughly 20% to 35%, driven by the compounding effects of higher penetration, increased temporal concentration of offshore wind output, and the structural inflexibility of lignite-based baseload capacity [29, 12, 14].

Whether cannibalisation is permanent or transient depends on the interplay between market structure and policy design. Theoretical models suggest convergence under idealised conditions [30, 31], but current European markets combine carbon pricing with direct VRE support mechanisms that decouple investment decisions from market signals [11], delaying or preventing the equilibrating exit of marginal generators that these models rely on. The gap between theory and practice is particularly stark in small, rapidly

expanding Baltic markets. A 2024 study of renewable expansion scenarios in Latvia and Estonia [32] projects wind revenue losses reaching 20 EUR/MWh in Latvia by 2029 and 15 EUR/MWh in Estonia under ambitious deployment targets, compared to near-zero losses today. The pace of planned expansion compresses the transition from low to high penetration into just a few years, leaving little time for the generation mix adjustments that the equilibrium models rely on.

2.3.2 Cross-border cannibalisation

Most analyses of cannibalisation have traditionally focused on individual national markets, treating each bidding zone as an isolated system. Stiewe, Xu, Eicke, and Hirth [12] challenged this assumption by providing the first comprehensive empirical estimate of cross-border cannibalisation spillovers. Using a panel dataset covering more than 30 European bidding zones over the period 2015 to 2023, they estimated a fixed-effects regression model in which the domestic wind value factor is regressed on both domestic wind penetration and the wind penetration of directly interconnected neighbours.

Their results reveal that cross-border effects are quantitatively significant: a one percentage point increase in domestic wind penetration reduces the domestic wind value factor by 0.61 percentage points, while a simultaneous one percentage point increase across all directly neighbouring markets reduces it by an additional 0.33 percentage points [12]. The combined effect is therefore nearly one full percentage point of value factor decline for every percentage point of coordinated European wind expansion. However, two features of their identification strategy warrant scrutiny. First, the use of annual data and zone fixed effects absorbs time-invariant market characteristics (generation mix, flexibility endowments) into intercepts rather than modelling them explicitly, limiting the framework's ability to explain *why* the same penetration increase produces different value factor declines in different markets. Second, wind penetration is not strictly exogenous, it responds to policy incentives that are themselves shaped by market prices, and while the fixed effects partially address this, they cannot eliminate reverse causality within zones over time. Despite these limitations, the finding carries a clear implication: the profitability of a wind farm in Denmark depends not only on Danish wind policy but also on the pace of deployment in Germany, the Netherlands, Norway, and Sweden.

Interconnection plays a dual role: transmission capacity allows surplus generation to be exported to neighbouring markets, relieving domestic price depression, but simultaneously imports the price-depressing effects of high wind generation in neighbouring zones. Stiewe et al. [12] finds that the mitigating effect of interconnection slightly outweighs the exacerbating effect, meaning well-interconnected bidding zones experience smaller declines in their wind value factor than isolated markets. Schlachtberger et al. [33] quantified this through a network optimisation model of the European electricity system, finding that transmission expansion across Europe reduces total system cost of a highly renewable grid by approximately 25%, largely by exploiting geographic diversity in wind resources.

The cross-border dimension is inescapable. The North Sea and Baltic Sea basins are connected through more than 30 cross-border interconnectors with a combined capacity exceeding 40 GW, linking Scandinavian, Central European, and Baltic markets [18]. Any assessment of wind energy value in one basin must therefore account for capacity expansion plans in the other. The Esbjerg Declaration's target of 150 GW of North Sea offshore wind by 2050 will not only depress capture values within North Sea bidding zones but will also spill over into Baltic markets through the Swedish, Finnish, and Polish interconnections that link both basins. In effect, no market is an island. Investment decisions in one basin are shaped by neighbours' expansion plans and policy choices. This interconnected reality is central to the thesis:

comparing the capture value potential of the North Sea and Baltic requires explicit modelling of cross-basin spillovers, not analysis of each basin in isolation.

2.4 Support Mechanisms and Revenue Risk

As capture values decline with rising penetration, the design of revenue support mechanisms becomes central to offshore wind investment viability. European governments have progressively shifted from guaranteed feed-in tariffs toward market-based instruments that expose generators to greater price risk while providing structured risk management. Table 2.1 summarises the principal instruments currently in use.

Table 2.1: Principal revenue support mechanisms for offshore wind energy. Market exposure ranges from zero (fixed feed-in tariff) to full (merchant), with two-sided contracts for difference (CfDs) dominating North Sea procurement and certificate or premium schemes more common in the Baltic. Basin prevalence reflects 2024 policy landscape.

Mechanism	**Description**	**Market exposure**	**Basin prevalence**
Feed-in tariff (FiT)	Fixed payment per MWh, independent of market price	None	Legacy (phased out)
Two-sided CfD	Strike price agreed; government pays shortfall, generator returns surplus	Medium	North Sea (UK, NL, DE, PL)
Electricity certificates	Tradable credits per MWh of renewable generation; value set by certificate market	High	Baltic (SE)
Premium-based schemes	Fixed or sliding premium added to the wholesale price	Medium–High	Baltic states (LT, LV, EE)
Subsidy-free / merchant	No direct financial support; revenue depends entirely on wholesale prices	Full	Emerging (NL, DE)

The contract for difference (CfD) has emerged as the dominant instrument. Under a two-sided CfD, the government and the generator agree on a strike price: when the wholesale price falls below the strike, the government pays the difference; when it rises above, the generator returns the surplus. This revenue stabilisation enables longer debt tenors and lower risk margins [34], with two-sided CfDs supporting 15 to 27 percentage points higher debt sizing compared to pure merchant exposure [35]. Kitzing et al. [36] argue that revenue stabilisation through CfDs is not merely a financial convenience but a structural necessity: without it, the cost of capital for offshore wind rises to levels that negate the LCOE reductions achieved over the past decade. In practice, CfD strike prices are determined through sealed-bid auctions where projects compete on price, introducing cost discipline but also deployment risk if bids are misjudged [37]. Electricity certificate schemes create additional price uncertainty: the value of tradable credits fluctuates based on supply and demand, exposing generators to commodity market risk that CfDs shield against [36]. Premium-based schemes occupy a middle ground, adding a fixed or sliding payment to the wholesale price, which exposes operators to the full downside risk of cannibalisation-driven price collapse during hours of high wind generation [35]. These design differences have material consequences for feasibility in low-value capture regimes.

The geography of support mechanisms differs between the two basins. North Sea markets have converged on mature CfD frameworks: the UK pioneered the instrument, the Netherlands conducts regular SDE++ tenders, and Germany has recently transitioned from negative bidding to CfD-based procurement [38]. Baltic markets utilise a more heterogeneous set of instruments: Sweden relies on technology-neutral

electricity certificates, Poland has adopted CfD auctions for its 5.9 GW programme [39], and the Baltic states use various premium-based schemes. Anatolitis et al. [37] show that auction design choices have first-order effects on realised build rates and bid prices, suggesting that this fragmentation may produce different deployment outcomes across the Baltic rim even where underlying resource quality is comparable. Whether subsidy-free offshore wind is viable in either basin remains contested: Jansen et al. [40] found that zero-subsidy bids in the Netherlands and Germany reflected specific market conditions that have since reversed. The EU Electricity Market Design reform of 2024 [10] mandated two-sided CfDs for all new publicly supported renewable capacity, establishing a pan-European floor for revenue stabilisation.

The commercial urgency is clear. The Boston Consulting Group [41] projects that German offshore wind projects face potential negative margins through 2035 under central scenarios, as capture prices fall towards or below LCOE [41, 42]. The geographic variation in mitigation levers (firming, forecasting, premium offtake) underscores a central point: the value of a wind farm depends not only on where the wind blows but on the market characteristics of the zone into which it sells.

2.5 Section Summary

Wind energy's market value is governed not by how much electricity a turbine produces but by the price at which it sells. The merit order effect depresses wholesale prices during high-generation hours; spatial correlation amplifies this across co-located assets; and interconnection propagates the effect across borders. The severity of the resulting cannibalisation varies substantially between markets depending on generation mix, system flexibility, and support mechanism design. These regional differences raise a commercially urgent question: what structural and geographic factors cause the same turbine technology to generate markedly different economic returns depending on where it is installed? This thesis addresses this question by developing a model that explicitly compares capture values across the North Sea and Baltic basins, accounting for each basin's distinct market structure, support mechanisms, and cross-border spillover dynamics.

3 Drivers of Regional Differences in Wind Value

The previous section established that the market value of wind energy declines with penetration and that this decline propagates across interconnected markets. Yet the rate and severity of this decline are far from uniform across Europe. A wind farm in Sweden earns a fundamentally different return per megawatt-hour than one in Germany, even when both face similar wholesale price levels, because the underlying market structures, flexibility resources, and wind generation patterns differ profoundly. This section examines three principal drivers of regional variation: the physical wind resource and its spatial characteristics, the architecture and composition of electricity markets, and the level of existing wind penetration, which determines where each market sits on the cannibalisation curve.

3.1 Wind Resources and Spatial Patterns

3.1.1 Atmospheric Circulation and Regional Wind Climates

European wind energy is governed by the large-scale westerly circulation that carries Atlantic weather systems eastward across the continent. Recurrent weather regimes modulate this flow. Grams et al. [43] identified seven such regimes through clustering of sea-level pressure fields: three cyclonic regimes that

collectively drive strong westerly flow on 38.2% of winter days, and four blocked regimes (38.8%) that stall high-pressure systems over Scandinavia or Central Europe. European mean wind generation swings from 22 GW during blocking to 44 GW during the Atlantic Trough, a volatility band of 66% of the winter mean [43]. The North Sea and Baltic respond in near-lockstep to these large-scale patterns, but their response is not identical: the Baltic's partial shielding by the Scandinavian land mass creates a degree of decorrelation that is relevant for portfolio diversification between the two basins.

Karagali et al. [44], using satellite-derived wind speed measurements from the pre-2013 period, characterised mean wind speeds of approximately 9.75 $\mathrm{m\,s^{-1}}$ over the North Sea compared with 7.5 to 8.0 $\mathrm{m\,s^{-1}}$ over the Baltic. The North Sea benefits from direct exposure to the Atlantic fetch, with dominant south-westerly to westerly winds during winter; the Baltic sits in a transitional zone between maritime and continental influences, with more varied wind directions and greater seasonal amplitude [44]. These satellite-derived estimates predate the rapid expansion of North Sea offshore capacity and do not capture the cluster wake effects that now reduce effective wind speeds in densely developed areas such as the German Bight, a limitation that complicates direct comparison with the sparser Baltic.

3.1.2 Capacity Factors: North Sea versus Baltic

The difference in mean wind speeds translates directly into a difference in energy yield. Solbrekke and Sorteberg [45] developed NORA3-WP, a high-resolution (3 km) offshore wind power dataset covering the North Sea, Baltic Sea, Norwegian Sea, and Barents Sea for the period 1996 to 2019. Their results confirm a clear gradient: capacity factors of 40 to 50% across the central and northern North Sea, 35 to 40% in the Baltic, and 45 to 55% along the exposed Norwegian coastline [45].

These figures, however, do not account for inter-farm wake losses, an effect of growing importance as North Sea wind farms proliferate and one that the NORA3-WP dataset, ending before the post-2020 capacity surge, cannot capture. Schneemann et al. [46] analysed wake properties in the German Bight using large-eddy simulations. They found wind speed deficits of 2 to 2.5 $\mathrm{m\,s^{-1}}$ within farm clusters that can extend up to 70 km downstream under stable atmospheric conditions. Given the cubic relationship between wind speed and power ($P \propto v^3$), a 10% speed deficit translates to roughly a 27% reduction in power output. The German Bight, where multiple gigawatt-scale clusters now operate in close proximity, is particularly affected. The Baltic Sea, with sparser development and greater inter-farm spacing, is less susceptible to cluster wake effects. The gap in effective capacity factors between the two basins may therefore be narrower than raw wind speed data suggest.

3.1.3 Spatial and Temporal Correlation

The correlation structure of wind generation across space and time is a critical determinant of portfolio-level cannibalisation. When wind farms across a region produce simultaneously, they collectively depress the same set of market prices; when their output is decorrelated, the price impact is distributed across different hours, reducing the severity of cannibalisation for each individual asset. Whether the North Sea and Baltic Sea are sufficiently decorrelated that spreading investment across both basins reduces overall cannibalisation exposure is an open empirical question.

Wind power correlation decays exponentially with distance, a pattern documented from transcontinental [47] to European scales [48], but the decay is neither isotropic nor uniform across settings. Directionally, east–west wind farm pairs exhibit a systematic time lag reflecting the west-to-east propa-

gation of Atlantic fronts, so that a wind event in the North Sea precedes the corresponding Baltic event by several hours [49]. This directional dependence partially decorrelates the two basins even under the same synoptic pattern, and cross-basin correlations are consistently lower than within-basin correlations [48]. Importantly, correlation decay is slower offshore than onshore: Potisomporn et al. [50] found this in UK waters, implying that the effective decorrelation length in offshore settings may exceed land-based estimates.

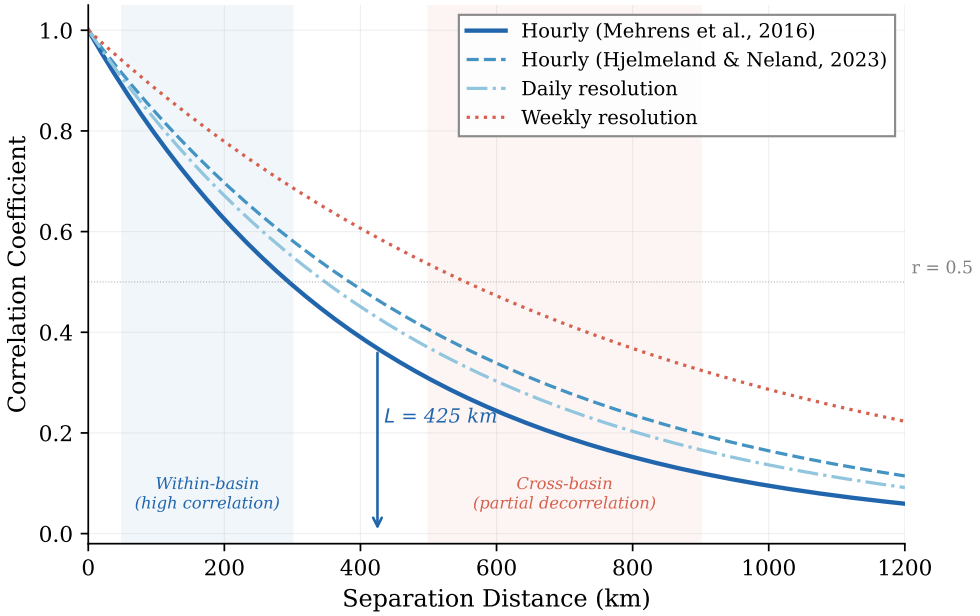


Figure 3.1: Exponential decay of wind power correlation with separation distance at different temporal resolutions, based on Mehrens et al. [51] and Hjelmeland and Neland [52]. Typical within-basin distances (50–300 km) exhibit high correlation, while cross-basin separations (500–900 km) show partial decorrelation. The decorrelation length $L \approx 425$ km marks the distance at which correlation falls to $1/e$.

Mehrens et al. [51] directly quantified the cross-basin correlation structure, finding a decorrelation length of approximately 425 km over the range 4 to 600 km (Figure 3.1). Given the North Sea's span of approximately 580 km in width and 960 km in length, and the Baltic's comparable lateral dimensions, this decorrelation length implies that individual wind farms within each basin are strongly correlated at hourly timescales; the German Bight, Dogger Bank, and the Danish North Sea all respond to the same Atlantic weather systems, meaning that high-generation events tend to be basin-wide. If the offshore decorrelation length exceeds the 425 km land-based estimate, as Potisomporn et al. [50] suggest, within-basin correlation among offshore farms may be stronger still.

Importantly, the rate of correlation decay depends on temporal resolution. Hjelmeland and Neland [52] showed that at hourly resolution the correlation coefficient decays to 0.5 at approximately 384 km, but at weekly and monthly aggregations correlations persist over much larger distances because weather systems span thousands of kilometres. Hourly correlations govern the severity of price depression during individual wind events, while longer-timescale correlations determine whether wind droughts affect both basins simultaneously.

Critically, while the spatial correlation of wind *generation* has been well characterised, the spatial correlation of electricity *prices* across these basins has not been systematically mapped in relation to wind output. The distinction matters: the SDAC market coupling mechanism transmits price effects across borders, so prices in physically distant but well-interconnected zones can be more correlated than their wind resources. Whether the partial decorrelation of wind output between the two basins translates into a corresponding decorrelation of capture prices, or whether market coupling erodes this diversification benefit, is an open empirical question with direct consequences for cross-basin investment allocation.

In synthesis, the North Sea offers a stronger wind resource but faces a critical constraint: its high spatial correlation concentrates wind-driven price depression into the same market hours basin-wide, compounding the cannibalisation effect. The Baltic offers a weaker resource, but partial decorrelation from the North Sea and sparser development create a temporal distribution of wind events, so high-generation hours are less perfectly synchronised. Whether this physical trade-off translates into an economic advantage depends entirely on the market structures into which each basin sells, examined in the following subsection.

3.2 Market Structure and Price Formation

3.2.1 European Electricity Market Architecture

The revenue that wind generators earn depends not only on the physical resource but on the design of the market in which they sell their output. The day-ahead market, typically accounting for 80 to 90% of merchant wind revenue [40], clears through the Single Day-Ahead Coupling (SDAC) mechanism, which employs the EUPHEMIA algorithm to simultaneously match orders across all participating European bidding zones [53]. Germany exemplifies this architecture in practice: when North Sea wind farms submit zero-bid offers into the German-Luxembourg (DE-LU) bidding zone, the SDAC algorithm simultaneously calculates the price impact in DE-LU, checks interconnection capacity to neighbouring zones (Netherlands, Denmark, France), and adjusts all zonal prices to reflect the marginal value of transmission. The price in any given zone reflects both domestic supply-demand conditions and the marginal value of interconnection with neighbours [53, 54]. This coupling is what transmits the merit order effect across borders, making the North Sea and Baltic markets interdependent in a way that national-level analyses cannot capture. For this thesis, the implication is direct: any model of wind capture value must treat both basins as a single coupled system rather than two independent markets.

3.2.2 Bidding Zones Around Each Basin

The geographic resolution at which prices are formed directly affects wind capture values. European electricity markets are divided into bidding zones, each clearing at a single price; Figure 3.2 provides a schematic overview across both basins. North Sea offshore wind sells into a set of large, relatively liquid bidding zones: Germany-Luxembourg (DE-LU), the Netherlands (NL), Belgium (BE), Denmark-West (DK1), and the five Norwegian zones (NO1 to NO5), along with the United Kingdom. Germany's single bidding zone, which masks significant north-south congestion, remains under review [55, 56]; a potential split would directly affect North Sea wind capture values by creating a low-price northern zone.

The Baltic basin is served by a more fragmented set of bidding zones that reflects both the Nordic tradition of zonal pricing and the legacy of post-Soviet electricity systems. The Nordic countries pioneered multi-zone market design: Sweden is divided into four zones (SE1 to SE4), Norway into five (NO1 to

NO5), and Denmark into two (DK1 and DK2). Fridolfsson and Tangerås [57] found that this multi-zone structure operates without systematic market power abuse, with transmission constraints creating natural price areas that reflect genuine physical congestion. Finland operates as a single zone, and Poland clears as one zone despite significant internal congestion. The three Baltic states (Lithuania, Latvia, and Estonia) each constitute their own bidding zone [18]. In February 2025, the three Baltic states completed synchronisation with the continental European grid, disconnecting from the Russian and Belarusian power system [58], this reshaped cross-border flow patterns and exposed Baltic prices more directly to Nordic and Central European market conditions [18].

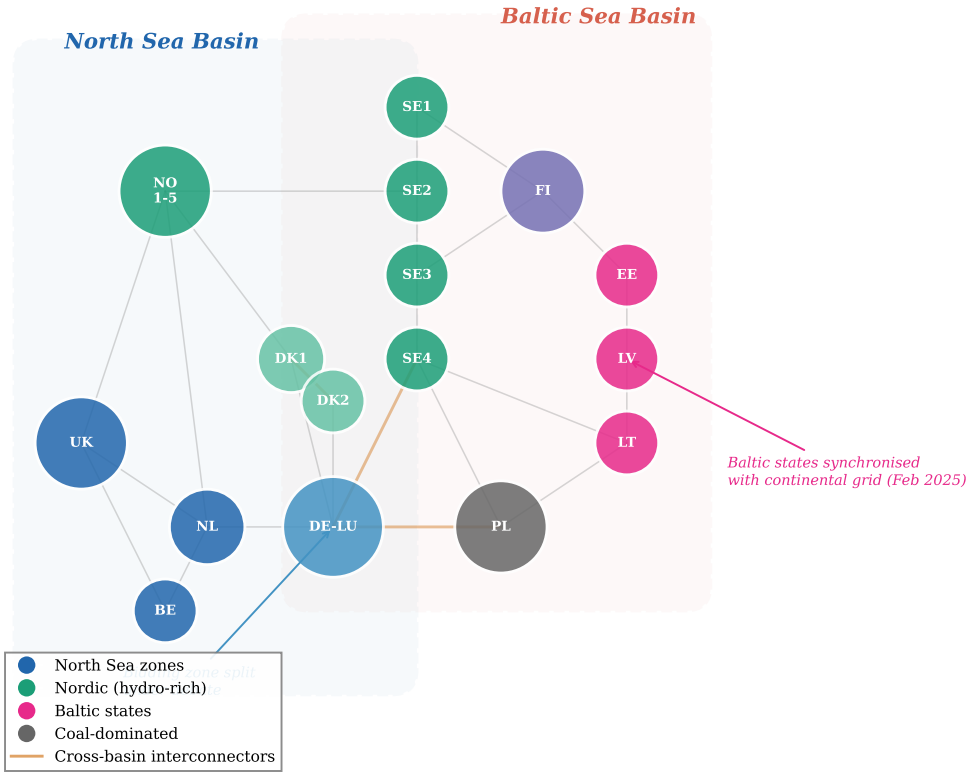


Figure 3.2: Schematic map of bidding zones and key interconnectors around the North Sea and Baltic Sea basins. Node colours indicate market archetype: blue for North Sea zones, green for hydro-rich Nordic zones, pink for Baltic states (synchronised with the continental grid in February 2025), and grey for coal-dominated systems. Orange lines highlight cross-basin interconnectors.

Cross-border transmission capacity determines the extent to which price differences can be arbitraged away. Flow-based capacity calculation in the Nordic region and the EU Clean Energy Package's requirement that member states make at least 70% of cross-zonal capacity available for trade are both pushing markets toward greater price convergence [17, 18]. The ENTSO-E Offshore Network Development Plan envisions more than €400 billion of offshore transmission investment by 2050 to support 189 GW of combined North Sea and Baltic capacity [59]. For wind generators, greater convergence is double-edged: surplus generation exports more easily, supporting capture values in the exporting zone, but importing zones absorb the corresponding price depression. The contrast in zonal architecture between the two

basins matters for capture value analysis: the Baltic's fragmented pricing structure localises price signals, meaning that cannibalisation in one zone does not automatically propagate to neighbours unless interconnection is sufficient.

3.2.3 Generation Mix and System Flexibility

The composition of a country's generation fleet determines the shape of its merit order curve, which in turn governs how sensitively wholesale prices respond to changes in wind output. Figure 3.3 compares the generation mix across the key countries bordering each basin. Four archetypes illustrate the range of market environments that offshore wind encounters across the two basins: a large thermal system (Germany), a small but exceptionally interconnected system (Denmark), a hydro-dominated system (Sweden), and a coal-heavy emerging market (Poland).

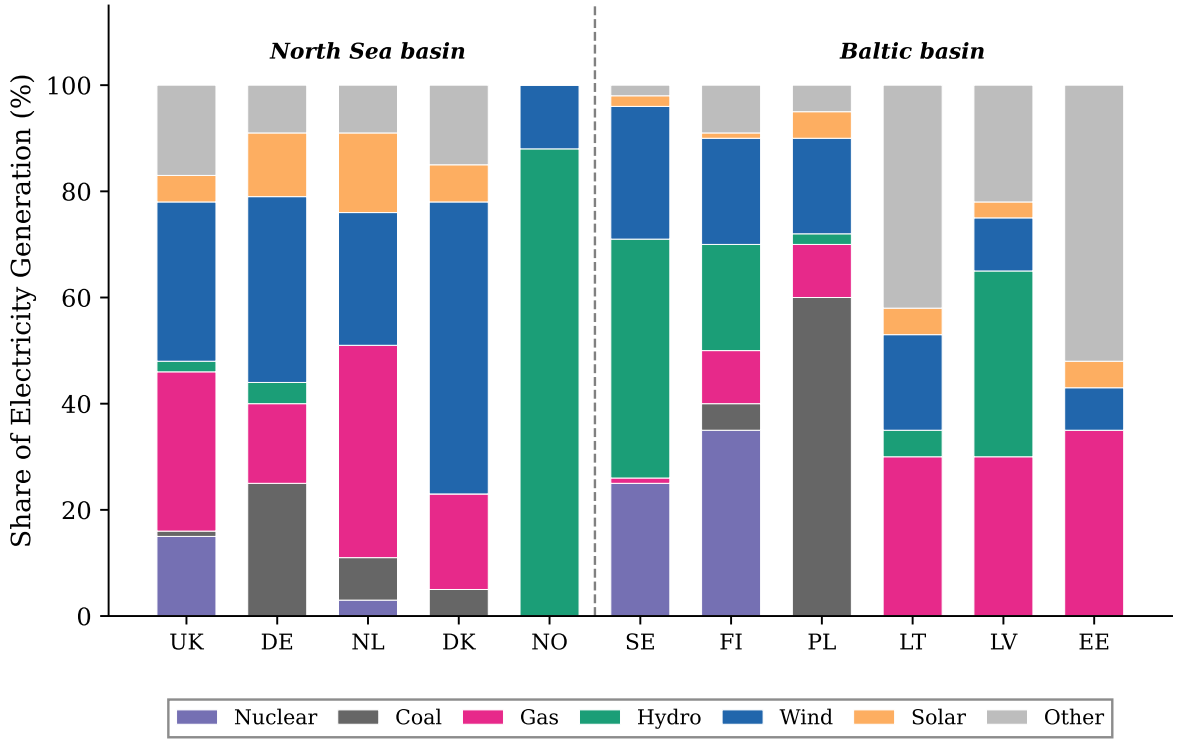


Figure 3.3: Electricity generation mix across North Sea and Baltic basin countries (approximate 2024 shares). Thermal-heavy systems (Germany, Poland) exhibit steep merit order curves where wind output produces large price swings; hydro-dominated systems (Sweden, Norway) absorb variability with minimal price impact. This diversity in generation portfolios is the primary structural driver of cross-basin differences in wind capture value. Data from Ember [14] and national transmission system operators.

Germany exemplifies the challenge facing thermal-heavy systems. With a dispatchable generation fleet still dominated by coal, lignite, and gas [14, 16], the German merit order is relatively steep, meaning that each additional unit of wind displacing a marginal gas plant produces a large price swing. When wind generation is high and pushes the clearing price down to the level of lignite must-run capacity or inflexible combined heat and power plants, the price drop is severe. Combined with Germany's position

as the largest European electricity market, this volatility propagates through SDAC to the Netherlands, Denmark, and beyond [60].

Denmark presents a contrasting case. Despite the highest wind penetration in Europe, approximately 55% of electricity generation [61, 14], Danish wind value factors have not collapsed as rapidly as theory would predict, because Denmark is exceptionally well interconnected relative to its size. Western Denmark has direct interconnections to Norway and Sweden, both of which offer large-scale hydropower storage that absorbs surplus Danish wind during windy periods and exports back during calm spells. This hydraulic flexibility effectively flattens the merit order curve that Danish wind generators face [11].

Sweden's hydro-dominated generation mix, approximately 45% of electricity production [14], provides the most favourable market environment for wind energy in the Baltic region. Hydropower reservoirs in northern Sweden act as a natural battery, storing potential energy during high wind and releasing it when wind falls. Hirth [11] estimated that at 30% wind penetration, the wind value factor is approximately 18% higher in hydro-dominated systems than in thermal-dominated systems, which is a gap large enough to shift a marginal project from loss-making to viable, and one that structurally favours Baltic offshore wind selling into Swedish zones over North Sea wind selling into German ones.

At the opposite end of the spectrum, Poland occupies a unique position on the Baltic rim. Its generation mix remains dominated by coal, still exceeding 60% in 2024 [16], with only nascent offshore wind development. Because wind energy displaces some of the most carbon-intensive and expensive generation in the European merit order, new offshore wind in Poland earns high capture values in the near term. Poland has planned 5.9 GW of Baltic offshore wind by 2030, a programme that would constitute the largest single addition of zero-marginal-cost capacity the Polish system has experienced [39]. As this capacity enters the market, it will progressively displace the coal-fired units that currently set marginal prices during high-demand hours, flattening the merit order and eroding the very conditions that make early projects attractive. Whether favourable capture values persist therefore depends on the pace at which coal retires relative to the pace at which offshore wind is added.

The contrast is sharp: an offshore wind farm in Sweden's SE4 zone and one in the Polish Baltic, separated by a few hundred kilometres, inhabit fundamentally different economic realities, with the former hydro-buffered and resilient to price depression, and the latter earning high capture prices today from displacing expensive coal but facing steep cannibalisation as 5.9 GW of new capacity enters a market with limited flexibility. Two further Baltic markets are entering the development cycle: Finland enacted its offshore wind act in January 2025 [62], enabling 6.5 GW by the late 2030s in a nuclear- and hydro-dominated system, while Estonia secured EU approval for a €2.6 billion support scheme in December 2024 [63]. Both sit at the very beginning of the penetration curve, offering near-zero cannibalisation today but facing rapid transitions as planned capacity materialises.

Table 3.1 consolidates the key characteristics of each market archetype discussed above. The cross-basin comparison becomes visible at once: the wind value factors that emerge are starkly different.

The pattern the table reveals is striking: every Baltic market records a higher value factor than every North Sea market, despite uniformly lower interconnection and, in Poland's case, near-zero system flexibility. Penetration level, not flexibility access, appears to be the dominant predictor of current value factor levels. Whether this ordering persists as Baltic capacity scales from single-digit to double-digit gigawatts is the question that a cross-basin comparison must answer.

Table 3.1: Summary of market archetypes across the North Sea and Baltic Sea basins. Wind penetration and value factor estimates reflect approximate 2024 values. The final column indicates the dominant revenue support instrument for offshore wind.

Country	Basin	Wind (%)	Dominant gen.	Flexibility	Intercon.	Value factor	Support
Germany	NS	∼35	Gas, coal, wind	Limited; gas peakers	Medium	0.76–0.85	Two-sided CfD
Denmark	NS/BS	∼55	Wind, gas	Nordic hydro (import)	Very high	0.80–0.90	CfD (from 2025)
Netherlands	NS	∼25	Gas, wind	Gas peakers; import	High	0.85–0.90	SDE++ tender
UK	NS	∼30	Gas, wind, nuclear	Gas CCGT; storage	Medium	0.80–0.85	CfD (AR rounds)
Sweden	BS	∼25	Hydro, nuclear, wind	Large hydro reservoirs	High	0.90–0.95	Elcert. (phasing out)
Finland	BS	∼20	Nuclear, hydro, wind	Hydro; nuclear baseload	High	0.90–0.95	Premium / CfD
Poland	BS	∼18	Coal, gas	Limited; coal inflexible	Low	0.95–1.00	CfD auction
Baltic states	BS	<15	Gas, import, wind	Limited; import-dep.	Medium	∼1.00	Premium schemes

3.2.4 Price Levels and Convergence Across Basins

For wind generators, absolute price levels matter less than the relationship between wind output and prices. German wholesale prices averaged above 50 €/MWh through 2022 and 2023, while Nordic zones traded at significantly lower levels [60, 18]. But a high average wholesale price is of limited value if capture prices fall substantially below it. The critical distinction between the two basins is their sensitivity to wind generation: the same wind event produces a deeper price depression in the German Bight than in the Swedish Baltic, even when the volume of generation is comparable. These price sensitivities have not been systematically mapped across the full set of bidding zones surrounding both basins.

Market structure and generation mix create a fundamental bifurcation in how the two basins absorb wind variability. Hydro-rich Nordic systems (Sweden, Norway, Finland) possess the flexibility to absorb concentrated wind output with minimal price impact: during high wind hours, water is stored instead of spilled, and during calm periods stored energy is released, flattening the merit order curve. Thermal-heavy systems (Germany, Poland) lack this buffer: when wind is high, inflexible coal and nuclear generation must be kept online or ramped down at cost, creating an abrupt price collapse that depresses capture values for all generators, not just wind. This structural difference means that two wind farms with identical generation profiles would earn fundamentally different revenues depending on which market they sell into. North Sea wind entering thermally-dominated zones (Germany, Benelux) experiences steep capture value degradation; Baltic wind selling into hydro-rich zones retains value. The question for cross-basin investment is whether superior North Sea wind resources compensate for less favourable market absorption, or whether market structure advantages in the Baltic outweigh lower physical resource quality.

3.3 The Penetration Gradient

A third factor mediates the interaction between wind resources and market structure: the existing level of wind penetration. Where a market sits on the cannibalisation curve determines the marginal impact of each additional gigawatt.

North Sea markets are already mature wind markets. Denmark generates approximately 55% of its electricity from wind [61], Germany approximately 35%, the Netherlands approximately 25%, and the

United Kingdom approximately 30% [14, 64]. These markets have experienced significant value factor erosion over the past decade. Baltic markets, with the partial exception of Sweden (approximately 25% and heavily buffered by hydro), are earlier on the penetration curve: Finland at approximately 20%, Poland at approximately 18%, and the Baltic states below 15% [14, 16].

Both basins have committed to dramatic expansion, underpinned by the broader EU targets set out in REPowerEU [6]. The Esbjerg Declaration of May 2022 pledged at least 65 GW of combined North Sea offshore wind by 2030 and 150 GW by 2050 [65]. Three months later, the Marienborg Declaration committed eight Baltic rim countries to at least 19.6 GW of Baltic offshore wind by 2030, a sevenfold increase from the 2.8 GW installed at the time [66]. Barzehkar et al. [67] estimate 93 GW of technical potential in the Baltic basin through GIS-integrated LCOE modelling, suggesting that the Marienborg target represents only a fraction of what is physically and economically viable. The ENTSO-E Offshore Network Development Plan consolidates these ambitions into a system-wide projection of 146 GW in the North Sea and 43 GW in the Baltic by 2050 [59].

As both basins expand simultaneously, cannibalisation will intensify across all markets, but the timing and severity will differ. North Sea markets, already at high penetration, face the steepest marginal declines in capture value. Baltic markets are earlier on the curve and, in the case of Poland and the Baltic states, stand to benefit from displacing expensive thermal generation. The question that this divergence raises is fundamentally a spatial and temporal forecasting problem: given the planned trajectories of expansion in each basin, what will capture values look like in 2030 and beyond? Answering this requires the modelling tools examined in the following section.

3.4 Section Summary

Three factors interact to create fundamentally different wind value environments around the North Sea and Baltic Sea: the wind resource (stronger but more spatially correlated in the North Sea; weaker but partially decorrelated in the Baltic), the market structure (hydro-dominated Nordic systems absorb variability with minimal price impact; thermal-heavy systems amplify it [11, 68]), and the level of existing penetration (mature North Sea markets already experiencing significant value erosion; Baltic markets in an earlier, more favourable phase). Yet the literature has not systematically compared capture values across these two basins under planned expansion scenarios.

4 Modelling Wind Value Across Space

This thesis must answer a question that no existing tool addresses in isolation: for each candidate offshore location across the North Sea and Baltic Sea, what hourly generation profile would a turbine produce, what price would that electricity command in the local wholesale market, and how does this revenue vary spatially in ways that current cost-based assessments miss? Answering it requires coupling three modelling traditions that have developed largely independently: wind resource modelling, which estimates when and how much electricity a turbine generates (Section 4.1); electricity price modelling, which quantifies how markets respond to variable generation (Section 4.2); and GIS-based spatial assessment, which maps physical and economic feasibility across space (Section 4.3). This section reviews each tradition, identifies the gap at their intersection, and sets out the methodological approach this thesis adopts to bridge it.

4.1 Wind Resource Modelling

4.1.1 Reanalysis Datasets

A comparative study spanning the North Sea and Baltic Sea requires a single reanalysis product applied with identical methodology across both basins. Table 4.1 compares the principal datasets available. The central trade-off is between spatial resolution and cross-basin coverage: higher-resolution regional products (NORA3 at 3 km, COSMO-REA6 at 6 km) outperform ERA5 in representing coastal wind gradients [45, 69], but none spans both basins within a single consistent framework with the hourly time series needed for capture value computation. ERA5 (ECMWF), with 0.25° resolution and hourly coverage from 1940 to the present, is the only product that does. Validation against 77 tall towers worldwide [70, 71] and four North Sea platforms ($r = 0.85$ to 0.93 at 100 m; [72]) confirms its reliability for the broad spatial patterns this thesis requires, even if it sacrifices fine-scale coastal detail.

Table 4.1: Comparison of reanalysis and wind resource datasets for European offshore wind assessment. ERA5 is the only product spanning both basins with hourly resolution from 1940 to present, making it the default choice for cross-basin capture value analysis despite its coarser spatial grid.

Dataset	Spatial	Temporal	Coverage	Geographic scope	Limitation for this thesis
ERA5 [71]	0.25° (∼31 km)	Hourly	1940–present	Global	Underestimates strong winds offshore
NORA3 [45]	3 km	Hourly	1996–2019	North Sea, Baltic, Norwegian, Barents	Ends before post-2020 capacity surge
NEWA [73]	3 km	Climatological	30-year mean	All Europe	No hourly time series for price matching
COSMO-REA6 [69]	6 km	Hourly	1995–2019	Central Europe only	Does not cover full Baltic basin
EMHIRES [74]	National	Hourly	1986–2015	EU member states	No site-level spatial resolution

ERA5 does exhibit systematic offshore biases. Gandoin and Garza [75] showed that it underestimates strong wind speeds non-linearly, by approximately 15% at 15 $\mathrm{m\,s^{-1}}$ and up to 20% at 20 $\mathrm{m\,s^{-1}}$, due to excessive sea-surface drag parameterisation. Given the cubic relationship between wind speed and power, even modest upper-tail biases propagate into significant capacity factor errors. Their proposed correction improves extreme wind representation by approximately 10% [75].

This thesis therefore uses ERA5 with the Gandoin and Garza bias correction as its resource modelling foundation. Hourly temporal coverage from 1940 to present across both basins is non-negotiable for computing capture values: the wind-price correlation that drives cannibalisation operates at the hourly timescale, and forward projections under Esbjerg and Marienborg expansion scenarios require a dataset that extends beyond the 2019 endpoint of higher-resolution alternatives like NORA3 and COSMO-REA6. The cost is coarser spatial granularity (31 km vs 3–6 km), acceptable for the basin-scale comparison this thesis pursues.

4.1.2 The Virtual Wind Farm Methodology

Converting gridded wind speed fields into hourly power output requires a standardised conversion pipeline. Staffell and Pfenninger [70] established the benchmark methodology, implemented through the

Renewables.ninja platform and now widely adopted.

The pipeline consists of four steps: interpolation of reanalysis wind speeds to farm coordinates using local regression (LOESS); extrapolation from reference height to hub height (80 to 150 m) using a logarithmic wind profile; conversion through the manufacturer's power curve; and Gaussian smoothing to represent sub-grid variability [70]. Bias correction is critical: uncorrected MERRA-2 underestimates observed onshore capacity factors by 26 to 32% and offshore by 12 to 15%. After applying a time-invariant multiplicative correction calibrated to national fleet data, estimates agree with observations to within 0.5 percentage points for most European countries [70].

The importance of this methodology for capture value assessment lies in its temporal fidelity. Hourly capacity factors at each location enable the computation of generation-weighted average prices through direct multiplication with hourly market prices. Any loss of temporal resolution would obscure the within-day correlation between wind output and price that drives cannibalisation. This pipeline forms the supply-side backbone of the present thesis: hourly wind power time series at each candidate location are the input from which capture values are ultimately derived.

4.2 Electricity Price Modelling

4.2.1 Taxonomy of Approaches

Electricity prices exhibit statistical properties that make them among the most challenging commodities to forecast: extreme volatility, seasonality at multiple timescales, mean-reversion punctuated by spikes, and an increasing incidence of negative values driven by renewable surplus [76]. The choice of modelling approach for this thesis is governed by a specific requirement: the model must quantify the *relationship* between wind generation and electricity prices across heterogeneous European markets, not simply forecast tomorrow's price in a single zone. Weron's [76] taxonomy classified electricity price forecasting methods into five families: multi-agent, fundamental, reduced-form, statistical, and computational intelligence models. Of these, fundamental models (which simulate the merit order clearing process) and statistical approaches are most relevant to wind value assessment, because the former can project capture values under hypothetical future scenarios while the latter can estimate empirical wind-price relationships [77, 78]. A critical distinction for this thesis is between forecasting models and structural models: the former excel at short-horizon price prediction in a single market, while the latter can explain cross-market variation in the wind-price relationship. Maciejowska [79] illustrated the value of structural approaches through quantile regression, showing that wind output compresses the upper tail of the price distribution more than it shifts the mean, explaining why capture prices decline faster than average prices as penetration increases.

A parallel tradition of energy system models, reviewed by Pfenninger et al. [80] and Ringkjøb et al. [81], highlights a persistent trade-off: tools with fine spatial resolution tend to simplify temporal dynamics, while those with rich temporal detail aggregate spatial granularity. For a study spanning over 30 bidding zones at hourly resolution, this trade-off becomes binding.

Table 4.2 compares the principal modelling tools relevant to this thesis, drawn from the taxonomies of Pfenninger et al. [80] and Ringkjøb et al. [81] and supplemented by recent developments.

Table 4.2: Comparison of modelling tools relevant to offshore wind value assessment. The final column identifies each tool's specific limitation relative to the integrated spatial-economic framework this thesis requires.

Tool	Type	Spatial	Temporal	Key strengths	Limitation for this thesis
atlite [82]	Resource conversion	0.05–0.25°	Hourly	Open-source; flexible weather data; land-use constraints	No market or revenue component
Renewables.ninja [70]	Resource conversion	Site-level	Hourly	Bias-corrected; validated globally; transparent	No market or revenue component
PyPSA-Eur [83]	System optimisation	Nodal	Hourly	Full European transmission; co-optimises dispatch and investment	Produces shadow prices, not observed market prices
EuroMod [77]	Dispatch / market	Zonal (27)	Hourly	Realistic price volatility; 40–43% error reduction	Requires commercial data; cannot project beyond calibration period
PLEXOS [84]	Unit commitment	Nodal/zonal	Sub-hourly	Industry standard; detailed generator constraints	Proprietary and costly; not reproducible
Stiewe et al. [12]	Panel regression	Zonal (30+)	Annual	Cross-border spillovers; robust to heterogeneity	Fixed effects absorb market characteristics; not spatially mapped
Bosch et al. [85, 86, 87]	GIS spatial	0.05°	32 slices/yr	Global coverage; depth, distance, LCOE	No market value or revenue dimension

The pattern across these tools is consistent: each excels along one dimension while neglecting another. Resource conversion tools (atlite, Renewables.ninja) produce spatially resolved hourly generation but contain no market component. System optimisation models (PyPSA-Eur, extended globally through the PyPSA-meets-Earth initiative [88]) capture network constraints but produce shadow prices, the dual variables of a cost minimisation problem, rather than the observed market prices that determine actual capture revenues. Dispatch models (EuroMod, PLEXOS) reproduce realistic price dynamics but require proprietary plant-level data that limits reproducibility. GIS frameworks (Bosch et al.) provide the geographic scaffold but lack temporal granularity and revenue modelling. Zeyringer et al. [89] demonstrated the practical consequence: power system designs ignoring the spatiotemporal coincidence of high wind and low prices overestimate renewable deployment potential. No existing tool integrates all three dimensions at the resolution this thesis requires: spatial resource, temporal price, and economic value.

4.2.2 The Wind-Price Relationship

The specific modelling challenge for this thesis is not price forecasting per se but the quantification of how wind generation affects electricity prices and, through that mechanism, determines the capture value at any given location. This relationship is the core of the cannibalisation problem: the negative covariance between wind output and wholesale price corresponds, in economic terms, to the profile cost defined in Section 2.1 [90, 91]. Modelling it requires a framework that (a) captures the nonlinear, market-specific sensitivity of prices to wind generation, (b) accounts for cross-border spillovers through interconnected bidding zones, and (c) can be applied consistently across markets with fundamentally different generation mixes and flexibility endowments.

Three classes of approach have been applied to this problem, each with distinct strengths and limitations. *Regression-based models* estimate the relationship empirically by regressing wholesale prices (or value factors) on wind penetration and control variables. Hirth [11] pioneered this approach by combining meta-regression across published studies with results from the EMMA dispatch model, estimating

the slope of the value factor decline as a function of wind market share. Stiewe et al. [12] extended it to a spatial panel framework with fixed effects across more than 30 European bidding zones, regressing domestic value factors on both domestic and neighbouring wind penetration while controlling for load, gas prices, and hydro availability. Their model specification takes the form:

$$VF_{i,t} = \alpha_i + \beta_1 \cdot WP_{i,t} + \beta_2 \cdot \overline{WP}_{N(i),t} + \gamma \mathbf{X}_{i,t} + \varepsilon_{i,t} \tag{4.1}$$

where $VF_{i,t}$ is the wind value factor in bidding zone i at time t, $WP_{i,t}$ is domestic wind penetration, $\overline{WP}_{N(i),t}$ is the capacity-weighted average wind penetration of directly interconnected neighbours, $\mathbf{X}_{i,t}$ is a vector of controls, and α_i are zone fixed effects. The estimated coefficients ($\hat{\beta}_1 = -0.61$, $\hat{\beta}_2 = -0.33$) quantify the domestic and cross-border cannibalisation effects respectively. The framework's strength lies in its ability to isolate the marginal effect of wind while controlling for confounders across 30+ markets. Its limitations are threefold. First, fixed effects absorb time-invariant market characteristics (generation mix, flexibility) into zone-specific intercepts rather than modelling them explicitly, preventing the framework from explaining *why* the same penetration increase produces different value factor declines across different markets. Second, annual temporal resolution smooths over within-year dynamics (seasonal wind patterns, fuel price spikes, interconnector congestion during storm events) that drive the most severe capture value erosion. Third, the linear specification imposes a constant marginal effect of penetration, whereas evidence reviewed in Section 2.2 suggests this effect accelerates at higher penetration levels.

López Prol, Steinbacher, and Zilberman [13] complemented this pan-European perspective with a study of cannibalisation dynamics across 24 European countries, confirming the nonlinearity that Cevik and Ninomiya [25] identified: the marginal price impact of each additional percentage point of wind accelerates at higher penetration levels. Their results suggest that the relationship between penetration and value factor is better described by a convex function than a linear one, an empirical regularity with direct implications for projecting capture values under the Esbjerg and Marienborg expansion scenarios.

Merit order simulation models take a structural approach, explicitly constructing the supply stack from individual generator costs and capacities. This enables scenario analysis because the stack can be modified to reflect future capacity changes or fuel price movements. Green and Vasilakos [92] applied this method to the UK market, estimating the merit order effect of offshore wind and finding that large-scale deployment would reduce wholesale prices but simultaneously erode the revenues of wind generators themselves. However, merit order simulation requires detailed plant-level data for every bidding zone in the analysis and assumes perfect competition that may not capture strategic bidding or the complex block-order structures of European day-ahead auctions [76, 93]. Mendes, Staffell, and Green [77] addressed some of these limitations with EuroMod, a dispatch model calibrated to 27 European countries that achieves 40 to 43% error reduction relative to linear supply curve models by incorporating non-convex generator constraints. EuroMod produces realistic price volatility including negative prices, but its calibration to a specific historical period limits its transferability to future market configurations.

Hybrid approaches combine elements of both traditions and increasingly incorporate flexibility technologies that could arrest the cannibalisation trajectory. Glaum, Stiewe, and Hirth [94] used a counterfactual dispatch model to estimate the price impact of offshore wind and hybrid projects across the coupled European day-ahead market, finding that hybrid configurations combining wind with electrolysis can reduce cannibalisation by absorbing surplus generation locally, a result with direct relevance to North Sea hydrogen strategies such as the Danish energy island programme. More broadly, battery storage and Power-to-X technologies represent the primary mechanisms through which future capture value declines

could be moderated: by shifting surplus generation out of low-price hours, they reduce the temporal concentration of output that drives the merit order effect. However, the literature has not yet integrated these flexibility pathways into spatial capture value models, meaning that projections of future value factor decline implicitly assume that today's limited flexibility endowments persist. Demand-side flexibility, including industrial load shifting and demand response programmes [90], represents a further mitigation channel that remains largely unquantified in the cannibalisation literature and falls outside the scope of this thesis. Van der Heijden et al. [95] confirmed the practical importance of cross-border integration: incorporating generation and price data from Germany, Norway, France, and Denmark reduced Dutch day-ahead price forecasting error by approximately 4%, reinforcing that any wind capture value model must account for conditions in neighbouring zones.

This thesis adopts a parameterised regression approach that extends Stiewe et al. by replacing zone fixed effects with explicit market parameters (generation mix composition, flexibility endowment, interconnection capacity). Three attributes justify this choice over alternatives: it explains cross-market variation rather than absorbing it into intercepts; it requires no proprietary plant-level data, unlike PLEXOS and EuroMod, ensuring reproducibility; and it can project forward under Esbjerg and Marienborg expansion scenarios, unlike calibrated dispatch models tied to a historical period. The cost is reduced short-horizon price accuracy relative to full structural models such as EuroMod, which achieves 40 to 43% error reduction through non-convex generator constraints [77]. For a thesis comparing capture values across two basins under future scenarios rather than forecasting next-day prices, this trade-off is justified.

4.2.3 Methodological Approach of This Thesis

The modelling gap this thesis addresses sits between the Stiewe et al. [12] panel regression and the Bosch-Staffell-Hawkes GIS framework [85, 86, 87]. The former quantifies how wind penetration affects value factors across European markets but does not map this relationship spatially; the latter maps where wind energy is physically and economically viable but ignores revenue.

This thesis addresses this gap by developing a wind-price model that extends the Stiewe et al. framework in three ways. First, rather than absorbing market characteristics into fixed effects, the model parameterises them explicitly: generation mix composition (thermal share, hydro share), interconnection capacity relative to peak load, and flexibility metrics enter as explanatory variables alongside wind penetration. This allows the model to explain cross-market variation in value factor sensitivity, not merely estimate an average effect. Second, the model incorporates the nonlinear penetration-value relationship documented by López Prol et al. [13] and Cevik and Ninomiya [25], enabling projections under expansion scenarios that push markets into penetration regimes beyond the historical sample. Third, the model is coupled spatially with the Virtual Wind Farm pipeline [70]: hourly generation profiles at each candidate offshore location are fed through the wind-price model to produce location-specific capture value estimates, which are then overlaid onto the GIS framework alongside capacity factor, water depth, distance to coast, and LCOE.

The result is a spatial assessment that ranks candidate sites not by how much energy they produce but by how much revenue they generate, integrating the supply-side tradition of Bosch et al. [85, 86, 87] with the market-side analysis of Stiewe et al. [12] within a single framework applicable to both basins. This approach necessarily inherits limitations from its constituent components (ERA5's offshore wind speed biases [75], the assumption that historical wind-price relationships extrapolate to higher penetration regimes, and the omission of future flexibility technologies) but it represents a step toward the integrated

spatial-economic assessment that the literature currently lacks.

4.3 GIS-Based Spatial Assessment

4.3.1 The Bosch, Staffell, and Hawkes Progression

The most comprehensive spatial framework for assessing wind energy potential at the global scale has been developed through a series of studies by Bosch, Staffell, and Hawkes. Bosch et al. [85] produced temporally resolved onshore wind capacity factors at a 0.05° grid using MERRA-2 data binned into 32 time slices per year. The 2018 extension [86] applied the methodology offshore with constraints for water depth and distance to coast, estimating global offshore potential at 390,000 to 570,000 GW. The 2019 study [87] added spatially resolved LCOE, showing that attractive sites cluster in shallow North Sea waters (£60 to 80/MWh) while remote deep-water sites exceed £200/MWh. Chu and Hawkes [96] extended the approach to 731 terrestrial and 339 offshore zones globally with hourly capacity factor profiles.

The framework's fundamental limitation is that it characterises costs comprehensively but ignores revenue entirely. A site with a low LCOE and a high capacity factor may nonetheless generate poor returns if it sells into a market where cannibalisation has depressed capture prices below the LCOE threshold. Even recent extensions toward hybrid offshore systems [94] treat capture value as an evaluation metric rather than a core spatial modelling output. This asymmetry, detailed cost modelling paired with absent revenue modelling, is the central gap that an integrated spatial-economic framework must fill.

4.4 Section Summary

Three modelling traditions have developed independently: wind resource modelling generates hourly generation profiles but contains no market component; electricity price modelling quantifies how wind penetration affects wholesale prices but does not map this spatially; and GIS-based frameworks characterise physical feasibility and costs but ignore revenue. Each excels along one dimension while neglecting the others.

Current energy system models sacrifice spatial resolution for temporal detail or vice versa; dispatch and market models require proprietary data that limits reproducibility; GIS frameworks omit the temporal dynamics and market integration that determine actual revenue. This thesis bridges the gap by developing a parameterised wind-price model, calibrated at the zonal level but applied spatially across candidate offshore sites, that explicitly incorporates market characteristics as explanatory variables rather than absorbing them into fixed effects.

This approach enables the cross-basin revenue-augmented spatial assessment set out in Section 5. Rather than ranking candidate sites by capacity factor or LCOE alone, it ranks them by capture value, integrating supply-side resource quality with demand-side price exposure through an empirically grounded framework applicable to both basins. The objectives that follow operationalise this framework.

5 Research Gap and Objectives

5.1 Identified Gaps

The preceding review has traced three interconnected lines of inquiry: the mechanisms that erode wind energy value, the regional factors that cause this erosion to vary, and the modelling tools available

to assess wind potential. Each is well developed in isolation; the problem is that they have not been brought together. Three gaps emerge at their intersection, each a consequence of the one before it.

Gap 1: No systematic comparison of wind energy value across the North Sea and Baltic Sea basins. Cannibalisation has been studied at the national level [11, 31, 13] and across interconnected European markets in aggregate [12, 25]. Jacquet [97] compared offshore wind capture values between the North Sea and the French Mediterranean, finding capture values projected to remain above 0.9 until 2050 in the Mediterranean, corresponding to revenues 10 to 39% higher than North Sea equivalents. However, the Baltic Sea was not examined. This omission is partly historical: until the 2022 Marienborg Declaration, the Baltic was not a major offshore wind basin, and most cannibalisation research followed capital allocation to the North Sea. It is also partly methodological: a meaningful cross-basin comparison requires coupling wind resource data with market-specific value models across fundamentally different market structures, which is a task that straddles the atmospheric science, energy economics, and power systems communities, each of which has developed its tools in relative isolation. This thesis extends Jacquet's comparative framework to the Baltic, testing whether its combination of hydro-rich Nordic markets, coal-heavy systems, and emerging markets creates a similarly favourable value environment, or whether the planned sevenfold expansion will erode that advantage before it can be captured.

Gap 2: GIS-based spatial assessment frameworks exclude market value. The most comprehensive spatial assessments [85, 86, 87, 96] characterise only the supply side: resource quality and construction cost. A location with a 45% capacity factor and an LCOE of €50/MWh appears attractive on these maps, yet if cannibalisation drives the capture price to €30/MWh, the project loses money. Conversely, a 35% capacity factor site in a market with robust capture values could yield higher returns. This gap persists because GIS-based frameworks originated in the renewable resource assessment community, where the relevant output is physical potential (GW, TWh), not revenue. Incorporating market value requires coupling spatial resource data with electricity market models that operate at a different resolution (bidding zones rather than grid cells) and a different disciplinary tradition (energy economics rather than geospatial engineering). No existing spatial model, including recent Baltic-specific work by Barzehkar et al. [67], has bridged this disciplinary boundary. Until it does, the cross-basin comparison identified in Gap 1 cannot be performed at the spatial resolution at which investment decisions are made.

Gap 3: No parameterised wind-price model that explains cross-market variation in cannibalisation severity. Incorporating capture value into a spatial framework requires a model that can estimate what a wind farm at a specific location will earn given the characteristics of the market it sells into. Hirth [11] estimated the wind-price relationship through meta-regression calibrated to individual countries; Stiewe et al. [12] extended this to 30+ bidding zones but absorbed market characteristics into zone fixed effects. The fixed-effects approach is a natural choice for panel econometrics, it controls for unobserved heterogeneity without requiring the researcher to specify what drives it, but the cost is that the model can estimate *that* different markets experience different rates of cannibalisation but not *why*. For this thesis, the distinction matters: a model that cannot explain cross-market variation cannot project capture values under future market configurations where the generation mix, flexibility endowment, and interconnection capacity differ from the historical sample. A parameterised model that accepts these characteristics as explicit inputs would enable two capabilities the literature lacks: forward-looking projections where historical fixed effects no longer apply, and spatial mapping of capture values by assigning each offshore grid cell to its nearest bidding zone. This model bridges Gaps 1 and 2, making both the cross-basin comparison and the revenue-augmented spatial assessment possible within a single framework.

5.2 Research Objectives

Addressing these three gaps requires working backwards from the spatial assessment (Gap 2) through the modelling tool it depends on (Gap 3) to the empirical comparison it enables (Gap 1). This thesis pursues five objectives, beginning with the foundational model and building outward to the spatial and comparative analyses it supports:

1. *Develop a parameterised wind-price model* (addressing Gap 3) that captures the relationship between wind penetration and capture value as a function of observable market characteristics: generation mix composition, interconnection capacity, and flexibility endowment. The model extends the Stiewe et al. [12] panel framework by replacing zone fixed effects with explicit market parameters, enabling application to bidding zones and scenarios beyond the historical sample.

2. *Map wind energy potentials* across the North Sea and Baltic Sea using ERA5-derived hourly capacity factors at each candidate offshore location, accounting for spatial correlation patterns, wake effects, and the bias corrections identified in the reanalysis literature [75, 70].

3. *Compare capture values across both basins* (addressing Gap 1) under current conditions and under the expansion scenarios defined by the Esbjerg (150 GW North Sea) and Marienborg (19.6 GW Baltic) Declarations, using the parameterised model to project how capture values evolve as each market moves along the cannibalisation curve.

4. *Integrate capture value into the GIS-based spatial assessment framework* (addressing Gap 2), overlaying revenue estimates onto the physical potential maps alongside capacity factor, water depth, distance to coast, and LCOE, and testing whether this additional dimension changes the ranking of candidate sites.

5. *Assess the diversification value* of spreading offshore wind investment across both basins rather than concentrating it in the North Sea, quantifying whether the Baltic's combination of lower correlation, hydro-buffered markets, and earlier position on the penetration curve translates into a measurable portfolio benefit for capture value.

Together, these objectives deliver a spatially explicit assessment of where offshore wind generates the most revenue, not merely the most energy. For an industry committing hundreds of billions of euros to North Sea and Baltic Sea capacity over the next two decades, the distinction between these two quantities will determine which projects survive without subsidy and which do not.

References

[1] Forschungsgesellschaft für Energiewirtschaft. German electricity prices at EPEX SPOT in 2024: Negative price hours reach record levels. FfE München, 2025.

[2] Nicoletta Colaluce. Europe offshore wind: Status and outlook 2025. WindEurope, 2025.

[3] European Commission. EU reference scenario 2020: Energy, transport and GHG emissions – trends to 2050. Technical report, European Commission, Directorate-General for Energy, 2021.

[4] Offshore WIND. Germany reaches 9.2 GW of offshore wind capacity. Offshore WIND, 2025.

[5] Baltic Energy Market Interconnection Plan. BEMIP offshore wind priorities and targets. European Commission, BEMIP High Level Group, 2024.

[6] European Commission. REPowerEU plan. Technical report, European Commission, 2022. COM(2022) 230 final.

[7] International Energy Agency. World energy outlook 2024. Technical report, IEA, 2024.

[8] International Renewable Energy Agency. World energy transitions outlook 2024: 1.5°C pathway. Technical report, IRENA, 2024.

[9] Mario Draghi. The future of European competitiveness: A competitiveness strategy for Europe. Technical report, European Commission, 2024. Report to the President of the European Commission.

[10] European Parliament and Council of the European Union. Directive (EU) 2024/1711 amending directive (EU) 2019/944 concerning common rules for the internal market for electricity. Technical report, Official Journal of the European Union, 2024. Adopted 13 June 2024; part of the EU Electricity Market Design reform package.

[11] Lion Hirth. The market value of variable renewables: The effect of solar wind power variability on their relative price. *Energy Economics*, 38:218–236, 2013.

[12] Clemens Stiewe, Alice Lixuan Xu, Anselm Eicke, and Lion Hirth. Cross-border cannibalization: Spillover effects of wind and solar energy on interconnected European electricity markets. *Energy Economics*, 143:108251, 2025.

[13] Javier López Prol, Andreas Steinbacher, and David Zilberman. The cannibalization effect of wind and solar in European electricity markets. *Resource and Energy Economics*, 76:101413, 2024.

[14] Ember. European electricity review 2025. Technical report, Ember, 2025.

[15] Tomasz Woźniak, Mariusz Skwarczynski, and Piotr Czarny. Price cannibalisation effects of wind energy in the Baltic Sea region: evidence from the Nordics and Poland. *Energy Policy*, 185:113956, 2024.

[16] Ember. Global electricity review 2024. Technical report, Ember, 2024.

[17] ACER. Key developments in European electricity and gas markets. Agency for the Cooperation of Energy Regulators, 2023.

[18] ACER. Key developments in European electricity and gas markets: 2025 market monitoring report. Technical report, Agency for the Cooperation of Energy Regulators, 2025.

[19] Frank Sensfuß, Mario Ragwitz, and Massimo Genoese. The merit-order effect: A detailed analysis of the price effect of renewable electricity generation on spot market prices in Germany. *Energy Policy*, 36(8):3076–3084, 2008.

[20] Johanna Cludius, Hauke Hermann, Felix Chr. Matthes, and Verena Graichen. The merit order effect of wind and photovoltaic electricity generation in Germany 2008–2016: Estimation and distributional implications. *Energy Economics*, 44:302–313, 2014.

[21] Petra Lunáčková, Jan Průša, and Karel Janda. The merit order effect of czech renewable energy. Technical Report CAMA Working Paper 2017-17, Crawford School of Public Policy, Australian National University, 2017.

[22] Klaas Würzburg, Xavier Labandeira, and Pedro Linares. Renewable generation and electricity prices: Taking stock and new evidence for Germany and Austria. *Energy Economics*, 40:S159–S171, 2013.

[23] Stefano Clò, Alessandra Cataldi, and Pietro Zoppoli. The merit-order effect in the Italian power market: The impact of solar and wind generation on national wholesale electricity prices. *Energy Policy*, 77:79–88, 2015.

[24] Sebastian Kolb, Marius Dillig, Thomas Plankenbühler, and Jürgen Karl. The impact of renewables on electricity prices in Germany — an update for the years 2014–2018. *Renewable and Sustainable Energy Reviews*, 134:110307, 2020.

[25] Serhan Cevik and Keitaro Ninomiya. Chasing the Sun and catching the Wind: Energy Transition and Electricity Prices in Europe. Technical Report WP/22/220, International Monetary Fund, 2022.

[26] Lion Hirth. The optimal share of variable renewables: How the variability of wind and solar power affects their welfare-optimal deployment. *The Energy Journal*, 36(1):149–184, 2015.

[27] Sashank Yamujala et al. Can Europe's decarbonisation goals offset the merit order effect and improve value factors of variable renewable energy towards 2050? *Energy Policy*, 211:115137, 2026.

[28] Oliver Ruhnau, Clemens Stiewe, Jarusch Müssel, and Lion Hirth. Natural gas savings in Germany during the 2022 energy crisis. *Nature Energy*, 8:621–628, 2023.

[29] Charles Monteil et al. European electricity review 2023: Cannibalisation and negative pricing trends. Technical report, ENTSO-E, 2023. Analysis of negative price hours and value factor erosion in European electricity markets.

[30] Werner Antweiler and Felix Muesgens. On the long-term merit order effect of renewable energies. *Energy Economics*, 99:105275, 2021.

[31] Tom Brown and Lina Reichenberg. Decreasing market value of variable renewables can be avoided by policy action. In *39th International Energy Workshop*, 2021. TU Berlin and Chalmers University of Technology.

[32] Eimantas Neniškis, Lina Galinis, and Vidas Lekavičius. Price Cannibalization Effect on Long-Term Electricity Prices and Profitability of Renewables in the Baltic States. *Sustainability*, 16(15), 2024.

[33] David P. Schlachtberger, Tom Brown, Stefan Schramm, and Martin Greiner. The benefits of cooperation in a highly renewable European electricity network. *Energy*, 134:469–481, 2017.

[34] Philipp Beiter, Julien Guillet, Malte Jansen, Emma Wilson, and Lena Kitzing. The enduring role of contracts for difference in risk management and market creation for renewables. *Nature Energy*, 9(1):20–26, 2024.

[35] Mak Đukan, Dogan Keles, and Lena Kitzing. The impact of two-sided contracts for difference on debt sizing for offshore wind farms. *The Energy Journal*, 46(5):145–188, 2025.

[36] Lena Kitzing, Malte Jansen, Philipp Beiter, et al. Facilitating the financing of offshore wind through revenue stabilisation. *Nature Energy*, 9:1–4, 2024.

[37] Vasilios Anatolitis, Pablo del Río, Lena Kitzing, Oscar Fitch-Roy, and László Szabó. Policy relevant lessons from research on renewable energy auctions. *Energy Policy*, 199:114452, 2025.

[38] WindEurope. Is negative bidding finally dead? WindEurope position paper, 2025.

[39] Grzegorz Owczarek and Adam Czyżak. Baltic Offshore Wind Energy Development—Poland's Public Policy Tools Analysis and the Geostrategic Implications. *Energies*, 14(16):4883, 2021.

[40] Malte Jansen, Iain Staffell, Lena Kitzing, Sylvain Quoilin, Edwin Wiggelinkhuizen, Bernard Bulder, et al. Offshore wind competitiveness in mature markets without subsidy. *Nature Energy*, 5:614–622, 2020.

[41] Boston Consulting Group. Commercial excellence: The untapped potential for renewable developers. Technical report, Boston Consulting Group, October 2025.

[42] International Renewable Energy Agency. Renewable power generation costs in 2024. Technical report, IRENA, 2025.

[43] Christian M. Grams, Remo Beerli, Stefan Pfenninger, Iain Staffell, and Heini Wernli. Balancing Europe's wind-power output through spatial deployment informed by weather regimes. *Nature Climate Change*, 7(8):557–562, 2017.

[44] Ioanna Karagali, Merete Badger, Andrea N. Hahmann, Alfredo Pe na, Charlotte B. Hasager, and Anna Maria Sempreviva. Spatial and temporal variability of winds in the Northern European Seas. *Renewable Energy*, 57:200–210, 2013.

[45] Ida Marie Solbrekke and Asgeir Sorteberg. NORA3-WP: A high-resolution offshore wind power dataset for the Baltic, North, Norwegian, and Barents Seas. *Scientific Data*, 9:362, 2022.

[46] Jens Schneemann, Frank Theuer, Andreas Rott, Martín Dorënkämper, and Bernhard Stoevesandt. Wake properties and power output of very large wind farms for different meteorological conditions and turbine spacings: a large-eddy simulation case study for the German Bight. *Wind Energy Science*, 7:2373–2391, 2022.

[47] Louie. Correlation and statistical characteristics of aggregate wind power in large transcontinental systems. *Wind Energy*, 17(6):793–810, 2014.

[48] Julius Olauson and Mikael Bergkvist. Correlation between wind power generation in the European countries. *Energy*, 114:663–670, 2016.

[49] A. Malvaldi, S. Weiss, D. Infield, J. Browell, P. Leahy, and A. M. Foley. A spatial and temporal correlation analysis of aggregate wind power in an ideally interconnected Europe. *Wind Energy*, 20(8):1315–1329, 2017.

[50] Panit Potisomporn and Christopher R. Vogel. Spatial and temporal variability characteristics of offshore wind energy in the United Kingdom. *Wind Energy*, 25(12):537–552, 2022.

[51] Axel Mehrens, Andrea N. Hahmann, Mikael Larsén, and Lueder von Bremen. Correlation and coherence of mesoscale wind speeds over the sea. *Quarterly Journal of the Royal Meteorological Society*, 142(700), 2016.

[52] Martin Hjelmeland and Jonas Kristiansen Neland. Correlation challenges for North Sea offshore wind power: a Norwegian case study. *Scientific Reports*, 13:18670, 2023.

[53] NEMO Committee. EUPHEMIA public description: Single price coupling algorithm. Technical report, 2020. PCR Project documentation.

[54] David Newbery, Michael G. Pollitt, Robert A. Ritz, and Wadim Strielkowski. Market design for a high-renewables European electricity system. *Renewable and Sustainable Energy Reviews*, 91:695–707, 2018.

[55] Jonas Egerer and Jörn Weibezahn. Electricity market effects of two price zones in the German electricity market. *Energy Economics*, 2015.

[56] ACER. Decision on the bidding zone review: Assessment of alternative bidding zone configurations in Germany. Technical report, Agency for the Cooperation of Energy Regulators, 2024. Bidding Zone Review process under Regulation (EU) 2019/943.

[57] Sven-Olof Fridolfsson and Thomas P. Tangerås. Market power in the Nordic electricity wholesale market: A survey of the empirical evidence. *Energy Policy*, 37(9):3681–3692, 2009.

[58] ENTSO-E. Synchronisation of the Baltic States with the Continental European Grid, 2025. Successful synchronization of Estonia, Latvia, and Lithuania with Continental Europe Synchronous Area on 9 February 2025.

[59] ENTSO-E. European offshore network transmission infrastructure needs: Pan-European summary. Technical report, ENTSO-E, 2024. Offshore Network Development Plan, TYNDP 2024.

[60] Georg Erdmann and Florian Zimmermann. European and German electricity prices in times of natural gas crisis. *Energy Policy*, 176:113497, 2023.

[61] Danish Energy Agency. Energy statistics 2024: Wind power in Denmark. Technical report, Danish Energy Agency, 2024.

[62] Finnish Government. Act on offshore wind power in the exclusive economic zone. Technical report, Ministry of Economic Affairs and Employment of Finland, 2025. Entered into force 1 January 2025; enables 6.5 GW target by late 2030s.

[63] European Commission. State aid: Commission approves €2.6 billion Estonian scheme to support offshore wind energy. European Commission press release, December 2024, 2024.

[64] WindEurope. Wind energy in Europe: 2024 statistics and the outlook for 2025–2030. Technical report, WindEurope, 2024.

[65] Government of Denmark and Government of Germany and Government of the Netherlands and Government of Belgium. The Esbjerg Declaration on the North Sea as a green power plant of Europe, 2022. Esbjerg, 18 May 2022.

[66] Baltic Sea Energy Security Summit. The declaration of energy ministers: The Baltic Sea energy security summit, 2022. Marienborg, 30 August 2022. Signed by Denmark, Estonia, Finland, Germany, Latvia, Lithuania, Poland, Sweden.

[67] Mojtaba Barzehkar, Kevin Parnell, Tarmo Soomere, and Marko Koivisto. Offshore wind power plant site selection in the Baltic Sea. *Regional Studies in Marine Science*, 73:103469, 2024.

[68] Jan Horst Keppler, Seb Phan, and Yannick Le Pen. The impacts of variable renewable production and market coupling on the convergence of French and German electricity prices. *The Energy Journal*, 37(3):343–359, 2016.

[69] Thomas Spangehl, Tim Hewson, and Stefano Grassi. Evaluation of high-resolution reanalyses for wind energy application in the North Sea and Baltic Sea. *Wind Energy Science*, 6(6):1537–1559, 2021.

[70] Iain Staffell and Stefan Pfenninger. Using bias-corrected reanalysis to simulate current and future wind power output. *Energy*, 114:1224–1239, 2016.

[71] Jaume Ramon, Llorenç Lledó, Verónica Torralba, Albert Soret, and Francisco J. Doblas-Reyes. What global reanalysis best represents near-surface winds? *Quarterly Journal of the Royal Meteorological Society*, 145(724):3236–3251, 2019.

[72] E.T.G. Bot, C.B.H. Eeckels, D.A.J. Wouters, et al. Offshore wind resource at the North Sea: Long-term measurement campaign and understanding wind conditions. Technical Report TNO 2024 R11678, TNO, 2024.

[73] Andrea N. Hahmann, Thomas Sile, Jake Badger, et al. The making of the New European Wind Atlas, part 1: Model suite. *Geoscientific Model Development*, 13(10):5053–5078, 2020.

[74] Iratxe González-Aparicio, Thomas Huld, Francesco Careri, Fabio Monforti-Ferrario, and Andreas Zucker. EMHIRES dataset: Part I: Wind power generation. Technical Report EUR 28171 EN, European Commission, Joint Research Centre, 2017.

[75] Rémi Gandoin and Jorge Garza. Underestimation of strong wind speeds offshore in ERA5: evidence, discussion and correction. *Wind Energy Science*, 9:1727–1745, 2024.

[76] Rafał Weron. Electricity price forecasting: A review of the state-of-the-art with a look into the future. *International Journal of Forecasting*, 30(4):1030–1081, 2014.

[77] Carla Mendes, Iain Staffell, and Richard Green. EuroMod: Modelling European power markets with improved price granularity. *Energy Economics*, 131:107343, 2024.

[78] Jesus Lago, Grzegorz Marcjasz, Bart De Schutter, and Rafał Weron. Forecasting day-ahead electricity prices: A review of state-of-the-art algorithms, best practices and an open-access benchmark. *Applied Energy*, 293:116983, 2021.

[79] Katarzyna Maciejowska. Assessing the impact of renewable energy sources on the electricity price level and variability – a quantile regression approach. *Energy Economics*, 85:104532, 2020.

[80] Stefan Pfenninger, Adam Hawkes, and James Keirstead. Energy systems modeling for twenty-first century energy challenges. *Renewable and Sustainable Energy Reviews*, 33:74–86, 2014.

[81] Hans-Kristian Ringkjøb, Peter M. Haugan, and Ida Marie Solbrekke. A review of modelling tools for energy and electricity systems with large shares of variable renewables. *Renewable and Sustainable Energy Reviews*, 96:440–459, 2018.

[82] Fabian Hofmann, Johannes Hampp, Fabian Neumann, Tom Brown, and Jonas Hörsch. atlite: A lightweight Python package for calculating renewable power potentials and time series. *Journal of Open Source Software*, 6(62):3294, 2021.

[83] Jonas Hörsch, Fabian Hofmann, David Schlachtberger, and Tom Brown. PyPSA-Eur: An open optimisation model of the European transmission system. *Energy Strategy Reviews*, 22:207–215, 2018.

[84] Energy Exemplar. PLEXOS integrated energy model. Energy Exemplar, 2024.

[85] Jonathan Bosch, Iain Staffell, and Adam D. Hawkes. Temporally-explicit and spatially-resolved global onshore wind energy potentials. *Energy*, 131:207–217, 2017.

[86] Jonathan Bosch, Iain Staffell, and Adam D. Hawkes. Temporally explicit and spatially resolved global offshore wind energy potentials. *Energy*, 163:766–781, 2018.

[87] Jonathan Bosch, Iain Staffell, and Adam D. Hawkes. Global levelised cost of electricity from offshore wind. *Energy*, 189:116357, 2019.

[88] PyPSA-Earth Contributors. PyPSA meets Earth: An Open Energy System Model Initiative. `https://pypsa-meets-earth.github.io/projects.html`, 2024. Accessed: 2025.

[89] Marianne Zeyringer, James Price, Birgit Fais, Pei-Hao Li, and Ed Sharp. Designing low-carbon power systems for Great Britain in 2050 that are robust to the spatiotemporal and inter-annual variability of weather. *Nature Energy*, 3(5):395–403, 2018.

[90] Lion Hirth, Falko Ueckerdt, and Ottmar Edenhofer. Integration costs revisited – an economic framework for wind and solar variability. *Renewable Energy*, 74:925–939, 2015.

[91] Falko Ueckerdt, Lion Hirth, Gunnar Luderer, and Ottmar Edenhofer. System LCOE: What are the costs of variable renewables? *Energy*, 63:61–75, 2013.

[92] Richard Green and Nicholas Vasilakos. The economics of offshore wind. *Energy Policy*, 39(2):496–502, 2011.

[93] Jacob Mays, David P. Morton, and Richard P. O'Neill. Investment effects of pricing schemes for non-convex markets. *European Journal of Operational Research*, 307(3):1413–1427, 2023.

[94] Philipp Glaum, Clemens Stiewe, and Lion Hirth. Offshore wind and hybrids: A counterfactual case study for impacts in the coupled day-ahead market of Europe. *Energy Economics*, 137:107791, 2024.

[95] Ties van der Heijden, Jesus Lago, Peter Palensky, and Edo Abraham. Electricity price forecasting in European day ahead markets: A greedy consideration of market integration. *IEEE Access*, 9:119954–119963, 2021.

[96] Cheng-Ta Chu and Adam D. Hawkes. A geographic information system-based global variable renewable potential assessment using spatially resolved simulation. *Energy*, 193:116530, 2020.

[97] Paul Jacquet. Modelling offshore wind energy potential and price cannibalisation in Northern and Southern Europe. Master's thesis, Imperial College London, 2024. Supervised by Malte Jansen, Arseniy Galimov, and Adam Hawkes.